

Fuzzy A* quantum multi-stage Q-learning artificial potential field for path planning of mobile robots

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ABSTRACT

Aiming at problem of the efficiency of mobile robots heavily depends on the employed path-planning algorithms that encounter significant challenges from dynamic obstacles and complex environments. This work develops a fuzzy A* quantum multi-stage Q-learning artificial potential field approach, which combines fuzzy A*, quantum multi-stage Q-learning, and artificial potential field (APF) algorithms. The fuzzy A* quantum multi-stage Q-learning APF algorithm combines a fuzzy system with an A* algorithm to improve the A* algorithm. In addition, the fuzzy A* quantum multi-stage Q-learning APF algorithm applies quantum computing methods and multi-stage training methods to improve the convergence speed of the Q-learning algorithm. If dynamic obstacles exist in environments and dynamic obstacles can block the global path, the obtained A* path points are adopted as sub-goal points for the APF to plan paths. If mobile robots fall into trap areas, the quantum multi-stage Q-learning is then invoked to plan a path between the local minimum trap and the sub-goal point. Except for escaping the local minimum trap, the proposed fuzzy A* quantum multi-stage Q-learning APF has the advantage of consuming less time. The study tests the fuzzy A* quantum multi-stage Q-learning APF on environments with narrow passages and traps. The results verify that the proposed fuzzy A* quantum multi-stage Q-learning APF can navigate beyond complex environment situations and break free from local minimum traps.

1. Introduction

The technology and application areas of mobile robots have developed continuously over the last decades (Seong et al., 2023). As mobile robots are gradually being applied in daily lives, the path planned by a mobile robot is put in high demand by realistic application requirements (Zhang et al., 2023a). Path planning, as an important area of the current research on mobile robots, aims to achieve autonomous obstacle avoidance (Wu et al., 2022) and to plan an optimal and collision-free path (Do et al., 2022). An outstanding path planned by path-planning algorithms has a positive impact on the dynamic performance of mobile robots (Xu et al., 2023).

Existing algorithms for path planning can be broadly classified into six categories, including graph search-based, sampling-based, optimization-based, reinforcement learning-based, local path-planning, and hybrid algorithms. Table 1 demonstrates the main features of several different classes of path planning algorithms.

The core idea of graph search-based path planning algorithms is that the algorithms decouple the workspace of mobile robots into a series of

simple raster regions (Keung et al., 2023). The graph search-based path planning algorithm finds the optimal path by searching roasters. The advantage of graph search-based path planning algorithms is that the computation of the algorithms is less and the length of planned paths is short (Andreychuk et al., 2022). However, graph search-based path planning algorithms have large bends at the inflection of paths. A forward search optimization algorithm can reduce unnecessary turns in paths (Akay and Yildirim, 2023). In addition, a multi-objective D* algorithm can reduce the distance and the bends of the paths (Li et al., 2023). However, the above path planning algorithm based on improved graph search still suffers from the disadvantage of a large degree of path bending.

The path planning process of local path planning algorithms focuses on the localization of environments to meet the needs of dynamic obstacle avoidance of mobile robots. Local path planning algorithms can adapt to rapid changes in environments (Guo et al., 2022). However, local path planning algorithms may fail to plan a path in complex environments because of a lack of global information (Yang et al., 2022). An improved artificial potential field (APF) algorithm that introduces

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Table 1
Six categories of path-planning algorithms.

Category	Algorithm	Major advantages	Major disadvantages
Graph search	Dijkstra; A*; D*	Suitable for complex environments	Need precise environmental information
Sampling-based	RRT; PRM	A 3D path can be planned	Fall into local optimum
Optimization-based	PSO; genetic algorithm; ant colony optimization	Suitable for complex environments	Time-consuming; Maybe not global optimum; Cannot avoid dynamic obstacles
Reinforcement learning	Q-learning	Suitable for complex environments; not need priori information	Not global optimum
Local path-planning	APF; dynamic window method	Avoid dynamic obstacles	Fall into local optimum
Hybrid algorithm	Proposed fuzzy A* quantum multi-stage Q-learning APF	Suitable for complex environments; escape from local minimum traps; avoid dynamic obstacles less consumption time	Maybe not the global optimum

velocity information and constructs eccentric ellipses can adapt to dynamic environments and predict changes in the relative distance between vehicles and obstacles (Huang et al., 2022). An improved APF algorithm that introduces localization accuracy forces can enhance underwater acoustic localization accuracy while ensuring effective obstacle avoidance (Sun et al., 2023a). However, the above improved algorithm based on local path planning algorithm still has the disadvantage of easily falling into local minima.

Reinforcement learning-based path planning algorithms can allow MRs to learn autonomously from environments and predict feasible paths (Yu et al., 2024). Reinforcement learning-based path planning algorithms enable mobile robots to autonomously plan optimal paths (Nugent et al., 2023). However, the effect of path planning of reinforcement learning-based algorithms depends on the effect of pre-training. An improved Q-Learning algorithm improves the efficiency of local path planning for mobile robots and avoids local optimal solutions by improving the Q-table initialization, action selection strategy and reward function (Zhou et al., 2024). An improved TD3 algorithm introduces the frame stacking technique and offline learning strategy can effectively improve the energy-efficient path planning capability of unmanned aerial vehicles in real-time environments (Hong et al., 2021). However, the above improved path planning algorithm based on reinforcement learning relies on the training effect of agents and cannot guarantee the effectiveness of the planning results.

In complex environments, the A* algorithm can plan an optimal solution quickly (Zhang et al., 2023b). The A* algorithm has heuristics and pruning strategies, thus avoiding searching all feasible paths and reducing the number of operations (Liu et al., 2022). The APF approach is simple in principle and easy to implement (Gao et al., 2023). Compared to non-local path planning algorithms, APF has a fast-real-time performance (Xie et al., 2022). The Q-learning can continuously explore and learn from the environment with limited information without prior knowledge (Low et al., 2022). Although path planning is a hot research area in mobile robots (Zhou and Ho, 2022), existing path-planning algorithms still have many deficiencies, as follows.

- 1) The path-planning algorithms lack convergences. A collision-free path can be planned when completeness path planning algorithms

plan a path, such as D* algorithm, which are sampling-based path planning algorithm (Chen et al., 2023). In addition, with enough running time, a collision-free path can be found when probabilistic complete path planning algorithms plan a path (Luis et al., 2022). Such as various swarm intelligence algorithms (Cui et al., 2024). In some cases, a collision-free path may not be found by partial path planning algorithms (Shin and Kim, 2021). For example, local path planning algorithms cannot plan a path when mobile robots are caught in a local minimum trap region (Li et al., 2022).

- 2) The planned path may not be optimal. For example, even though the Dijkstra-planned path can be considered the shortest path available in the current 2D raster environment (Dai et al., 2021), the planned path length depends on raster sizes (Sun et al., 2023b). If the raster size is reduced, the length of paths will be reduced correspondingly (Han and Koenig, 2022a). In addition, some algorithms may not be able to provide optimal paths because the computational process of some algorithms is probabilistic (Aslan, 2022), such as swarm intelligence algorithms and sampling-based algorithms (Huang et al., 2023a).
- 3) The computation time of path-planning algorithms may be excessively long. The time consumption of various algorithms is small and does not differ significantly for a simple environment (Huang et al., 2023b). However, time will be severely consumed in complex environments. For example, the Q-learning faces the curse of dimensionality for global complex environments (Liu et al., 2021).

Three major improvements of existing path-planning algorithms are as follows.

- 1) The structural improvements for individual path-planning algorithms. Combining a bidirectional exploration strategy with kinematic constraints into the rapidly exploring random tree (RRT) can reduce time consumption and unnecessary growth of random trees (Wang et al., 2021a). A heuristic function with directional information can increase convergence speed and reduce 22.2% turns in paths of ant colony optimization (Wu et al., 2023). The number of iterations and path nodes can be reduced by 18.11% and 23.02% respectively when the construction of a random tree is combined (Cheng et al., 2023). The artificial bee colony (ABC), which suffers from slow convergence, can be improved by integrating a framework with more autonomy and intelligence (Ha et al., 2022).
- 2) Multi-algorithm fusion for improving algorithm performance. The incorporation of the global optimal solution update strategy into the particle swarm optimization (PSO) results in diminished particle oscillations during an evolutionary process and increased convergence speed (Yu et al., 2022). In addition, the time consumption and the planned path length of the ABC combined with reinforcement learning are reduced (Cui et al., 2022). Furthermore, the RRT algorithms integrated with deep neural networks can predict regions where feasible paths may exist for speeding up convergence (Wang et al., 2021b). Besides, the PSO that combines search and swoop mechanisms in the bald eagle search strategy can reduce path length by 4.45% (Dian et al., 2022).
- 3) The collaborative planning of multiple mobile robots. The A* algorithm combined with the multi-objective APF enhances the planning performance of multi-mobile robot formations and avoids collisions between mobile robots (Tong et al., 2022). A multi-objective PSO that combines fuzzy inference can integrate the kinetic and incomplete constraints of vehicle-like mobile robots (Sathiya et al., 2022). An ABC- probabilistic roadmap (PRM)- evolutionary programming can enhance the effectiveness of the planning process (Sunil and Sikander, 2022). A multi-modal collaborative multi-objective PSO algorithm based on reinforcement learning can plan the optimal path for multi-mobile robots under a multi-constrained environment (Zhang et al., 2022).

Therefore, this study innovatively proposes the fuzzy A* quantum multi-stage Q-learning APF to comprehensively consider both global and local planning under complex environments. This proposed fuzzy A* quantum multi-stage Q-learning APF combines the fuzzy A* algorithm, quantum multi-stage Q-learning, and APF for three steps. The major features of the fuzzy A* quantum multi-stage Q-learning APF are as follows.

- 1) To the best knowledge of the authors, this work combines graph search, local path-planning, and reinforcement learning methods to comprehensively consider global and local planning under complex environments for the first time.
- 2) In the first step of the fuzzy A* quantum multi-stage Q-learning APF, the fuzzy A* algorithm plans a global path. After the global path is planned, redundant nodes from the global path are eliminated. If dynamic obstacles exist in environments and can block the global path, the remaining path points are utilized as sub-goal points in a local path-planning process in the next step.
- 3) The A* algorithm cannot avoid dynamic obstacles in complex environments. In contrast, the APF of fuzzy A* quantum multi-stage Q-learning APF subsequently executes a local path-planning process, plans a path to reach sub-goal points, and avoids dynamic obstacles.
- 4) When the APF-based mobile robot is unable to continue with path planning, quantum multi-stage Q-learning is implemented to further plan paths. Consequently, the fuzzy A* quantum multi-stage Q-learning APF balances global and local planning, escapes from local minimum trap regions, avoids dynamic obstacles, and achieves the final goals. Because of such a combination, the fuzzy A* quantum multi-stage Q-learning APF can rapidly plan a collision-free path.

The remaining parts of the study are divided into the following sections. In Section 2, this study discusses major challenges encountered by path-planning algorithms that are applied by mobile robots. In Section 3, this study introduces the proposed fuzzy A* quantum multi-stage Q-learning APF. Section 4 provides an analysis of the results of the fuzzy A* quantum multi-stage Q-learning APF. Finally, Section 5 concludes this study.

2. Path-planning problems

This study aims to address the problems of path planning that comprise obstacle avoidance, and global and local path planning.

2.1. Obstacle avoidance

The obstacle avoidance capability, which includes both static and dynamic obstacle avoidance, is the fundamental function of mobile robots (Jin and Er, 2022) (Fig. 1). The difficulty in obstacle avoidance is the avoidance of dynamic obstacles (Aslan and Erkin, 2023). Local path planning algorithms are generally applied to solve dynamic obstacle avoidance problems for mobile robots (Zhang et al., 2021). When a path is blocked by a dynamic obstacle, a significant number of computational resources is wasted by a global path-planning algorithm (Zhou et al., 2021). Therefore, path-planning algorithms that consider dynamic obstacles are essential.

2.2. Global and local path planning

Global path planning algorithms alone require precise information about obstacles to be known in advance and often cannot cope with dynamic obstacles in the environment (Han and Koenig, 2022b). Local path-planning algorithms alone are possible not to plan a collision-free path for the reason of inadequate global information (Wang et al., 2022). The case where global and local path planning is not reconciled is illustrated in Fig. 2. Specifically, Fig. 2(a) gives a scenario where the A* algorithm is unable to dynamically avoid obstacles. In Fig. 2(b)–a

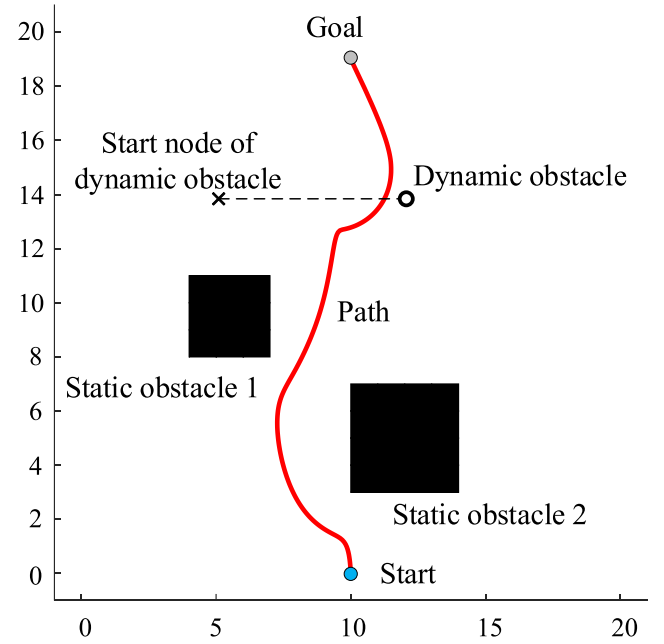


Fig. 1. Obstacle avoidance process for dynamic and static obstacles.

mobile robot based on APF moves in the wrong direction because of lacking global information.

2.3. Escaping from local minimum traps

In local minimum traps of APF, the first case is when a single obstacle blocks a mobile robot from moving, then the mobile robot may oscillate near this obstacle (Fig. 3(a)). Besides, a mobile robot may be trapped in a trap area when encountering a concave obstacle (Fig. 3(b)). Additionally, a mobile robot passing through narrow passages may be stuck in a local minimum trap (Fig. 3(c)). If a goal point is overly near obstacles, since the attractive force on mobile robots near the goal point is always less than the repulsive forces, the mobile robots cannot reach the destination (Fig. 3(d)).

3. Fuzzy A* quantum multi-stage Q-learning artificial potential field

The proposed fuzzy A* quantum multi-stage Q-learning APF combines fuzzy A*, APF, and quantum multi-stage Q-learning. The fuzzy A* algorithm aims to generate a series of global path points. The APF aims to plan a path between the global path points. The APF gives mobile robots the ability to avoid obstacles dynamically. The quantum multi-stage Q-learning-based mobile robots can escape from local minimum trap areas. Fig. 4 shows the workflow of the fuzzy A* quantum multi-stage Q-learning APF algorithm proposed in this study.

3.1. A* algorithm

The A* algorithm computes the heuristic function of nodes to obtain the optimal path. The heuristic cost $f(n)$ can be expressed as follows.

$$f(n) = \alpha_g g(n) + \alpha_h h(n), \quad (1)$$

where α_g is the coefficient of $g(n)$; $g(n)$ is the actual cost to reach node n from the start point; α_h is the coefficient of $h(n)$; $h(n)$ is the estimated cost to reach the goal from node n .

When $h(n)$ is less than the actual distance between the position of mobile robots and the goal, the path length calculated by the A* algorithm is short. However, a short-planned path is time-consuming for

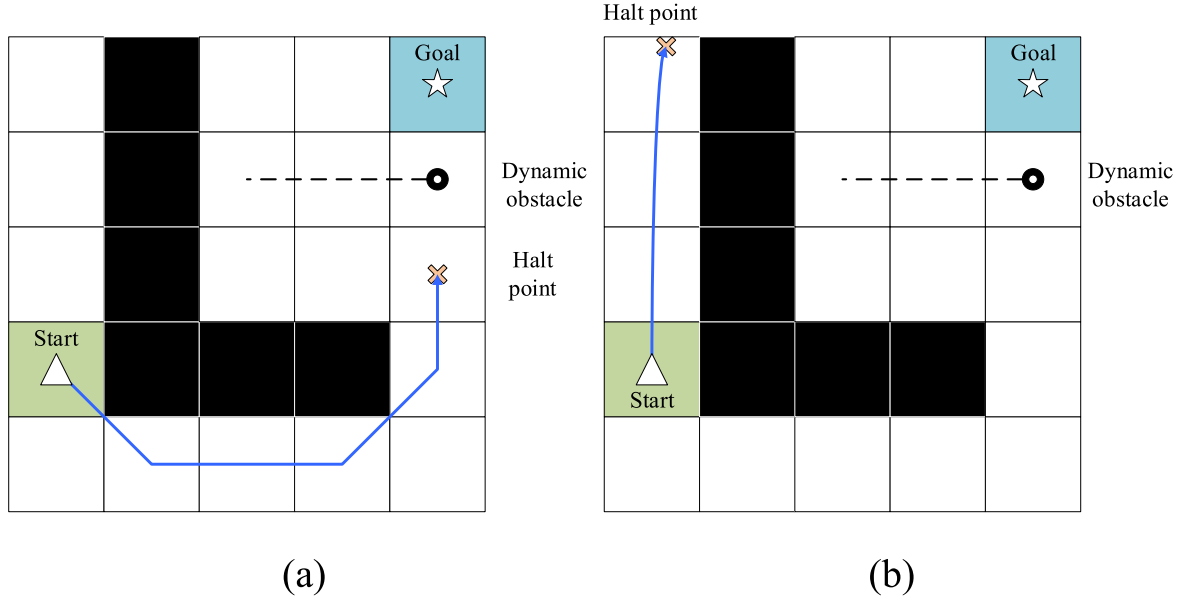


Fig. 2. Cases where global and local path planning is not reconciled: (a) A* algorithm cannot reach goal point; (b) APF cannot reach goal point.

planning algorithms. When $h(n)$ is larger than the actual distance between the position of mobile robots and the goal, the A* algorithm has a fast search speed at the cost of paths that tend to fall into local optimum solutions. The accuracy of the A* algorithm improves when $h(n)$ is closer to the shortest distance between the current and goal nodes. The A* algorithm can employ two distance calculation functions: the Manhattan and Euclidean distance functions. The Manhattan distance function $L_1(n)$ and the Euclidean distance function $L_2(n)$ are

$$L_1(n) = |n_x - e_x| + |n_y - e_y|, \quad (2)$$

$$L_2(n) = \sqrt{(n_x - e_x)^2 + (n_y - e_y)^2}, \quad (3)$$

where (n_x, n_y) is the current position of an agent; (e_x, e_y) is the position of the goal.

The A* algorithm commonly employs the eight- or four-neighbor expansion approach for neighborhood expansion. The benefits of the four-neighbor expansion are simplicity and avoiding a case where the path is overly close to obstacles. However, this four-neighbor extension produces a more tortuous path as the turn angle is limited to an integer multiple of 90° . The eight-neighbor expansion produces a path with a turn angle that is only an integer multiple of 45° . Although the eight-neighbor expansion-generated path is smoother than the four-neighbor expansion-generated path, the eight-neighbor expansion-generated path may be close to obstacles.

Fig. 5 depicts the general process of the A* algorithm. Specifically, the nodes of the A* algorithm are expanded toward the eight nodes adjacent to the start node. The nodes that meet the requirements are placed in an open-list table. The cost function of each node in the open-list table is calculated and compared by Eq. (1). A node with the smallest node cost function is applied as the next expanded node. Subsequently, the open-list table eliminates this node. The close-list table adds this node. The A* algorithm repeats the above process and ends when the nodes are expanded to the goal node.

3.2. Artificial potential field of fuzzy A* quantum multi-stage Q-learning artificial potential field

The APF is a virtual force-based approach applied for local path planning. This approach views the goal point as an attractive field source, which creates an attractive force for guiding mobile robots toward the destination. Moreover, the APF views an obstacle as a repulsive field source. The repulsive field source creates a repulsive force for mobile robots. The APF can avoid obstacles and arrive at the destination under the effect of combined forces.

The attractive potential field function $U_{att}(q_{agent})$ and the attractive force function $F_{att}(q_{agent})$ can be expressed as follows.

$$U_{att}(q_{agent}) = \frac{1}{2} K_a \rho^2(q_{agent}, q_{goal}), \quad (4)$$

$$F_{att}(q_{agent}) = -\nabla U_{att}(q_{agent}) = K_a \rho(q_{agent}, q_{goal}), \quad (5)$$

where K_a is the attractiveness gain coefficient; q_{agent} is the position of mobile robots; q_{goal} is the position of the goal; $\rho(q_{agent}, q_{goal})$ is the distance between the mobile robot and the goal point.

The representations for the repulsive potential field function $U_{rep}(q_{agent})$ and the repulsive force function $F_{rep}(q_{agent})$ can be expressed as follows.

$$U_{rep}(q_{agent}) = \begin{cases} \frac{1}{2} K_r \left(\frac{1}{\rho(q_{agent}, q_{obs})} - \frac{1}{\rho_d} \right)^2 & \rho(q_{agent}, q_{obs}) < \rho_d \\ 0 & \rho(q_{agent}, q_{obs}) \geq \rho_d \end{cases}, \quad (6)$$

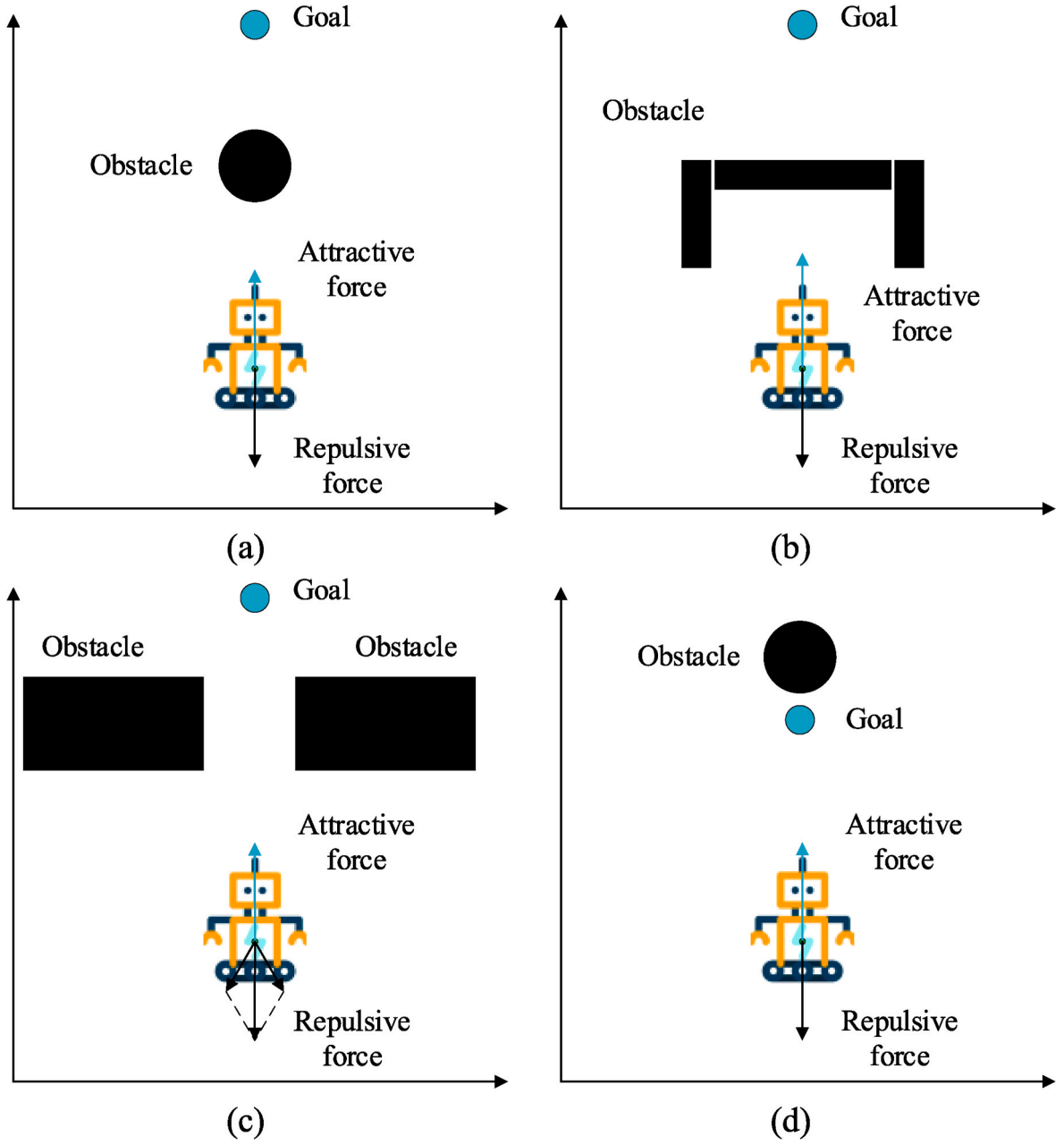


Fig. 3. Situations in which mobile robots fall into local minimum traps: (a) single obstacle exists between a mobile robot and the destination; (b) mobile robot enters a concave obstacle area; (c) mobile robot passes through narrow passage; (d) goal point is overly close to obstacle object.

$$F_{\text{rep}}(q_{\text{agent}}) = -\nabla U_{\text{rep}}(q_{\text{agent}}) = \begin{cases} K_r \left(\frac{1}{\rho(q_{\text{agent}}, q_{\text{obs}})} - \frac{1}{\rho_d} \right) \frac{1}{\rho(q_{\text{agent}}, q_{\text{obs}})^2} & \rho(q_{\text{agent}}, q_{\text{obs}}) < \rho_d \\ 0 & \rho(q_{\text{agent}}, q_{\text{obs}}) \geq \rho_d \end{cases}, \quad (7)$$

where K_r is the repulsive force gain coefficient; $\rho(q_{\text{agent}}, q_{\text{obs}})$ is the distance between the mobile robot and the obstacle; ρ_d is the threshold value of the distance.

The total force acting on a mobile robot is the sum of attractive and repulsive forces can be expressed as follows.

$$F(q_{\text{agent}}) = F_{\text{att}}(q_{\text{agent}}) + F_{\text{rep}}(q_{\text{agent}}), \quad (8)$$

where $F(q_{\text{agent}})$ is the combined force on the moving mobile robot.

The forces of APF-based mobile robots are shown in Fig. 6. The APF has the advantage of being easy to implement and enables dynamic

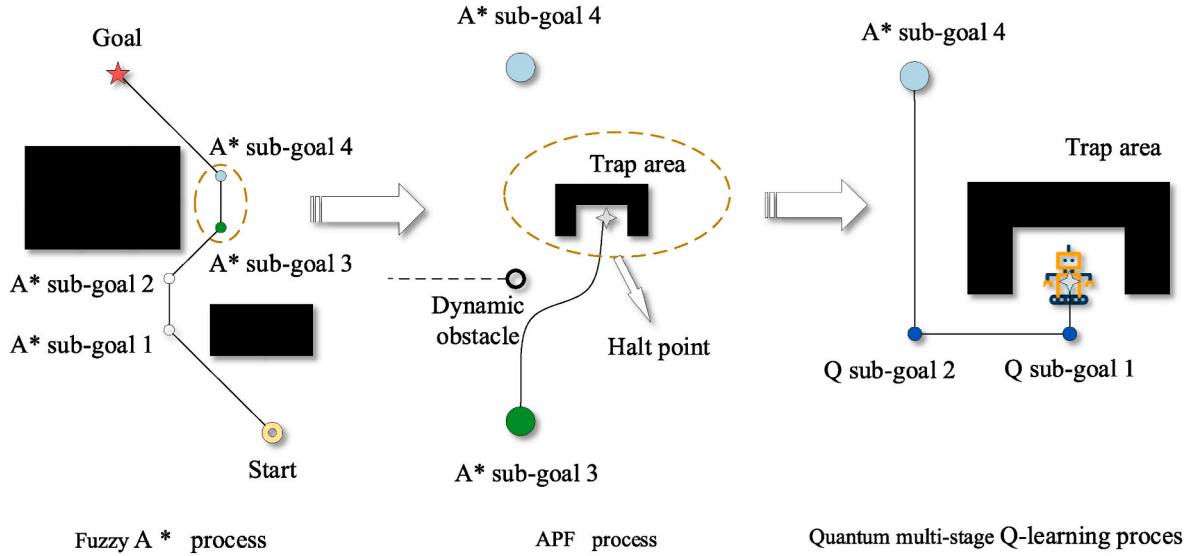


Fig. 4. Workflow of fuzzy A* quantum multi-stage Q-learning APF.

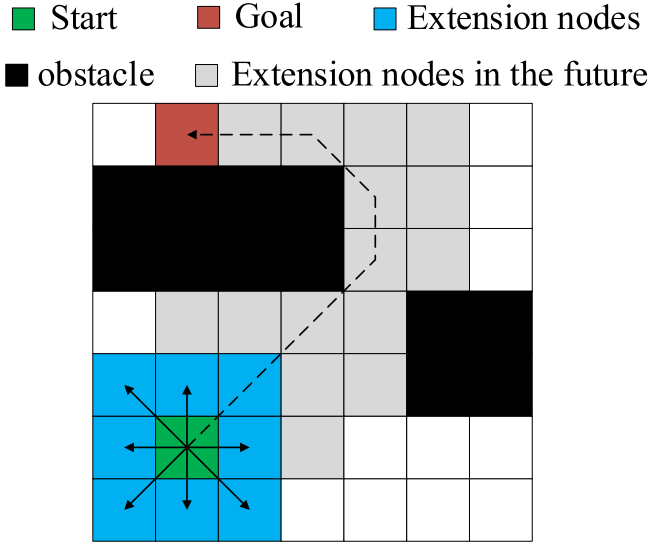


Fig. 5. A* algorithm planned path.

obstacle avoidance. However, as an APF-based mobile robot is affected by various obstacles, the mobile robot may get caught up in a trap area. If the distance from the mobile robot to the target point is overly large, the mobile robot may be subjected to excessive attractive forces causing the mobile robot to crash into obstacles.

3.3. Q-learning

Q-learning, as a machine learning method, can learn from environments with little or no information. The Q-learning can be described by a Markovian framework (Tsitsiklis, 1994). The components of Q-learning involve the mobile robot (agent), state, action, environment, and reward. The Q-learning self-learning process is achieved through the continuous exchange of information between mobile robots and environments. The behavioral value function is updated as follows.

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha_L [R_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)], \quad (9)$$

where s_t is current state; s_{t+1} is the next state; R_t is the immediate return

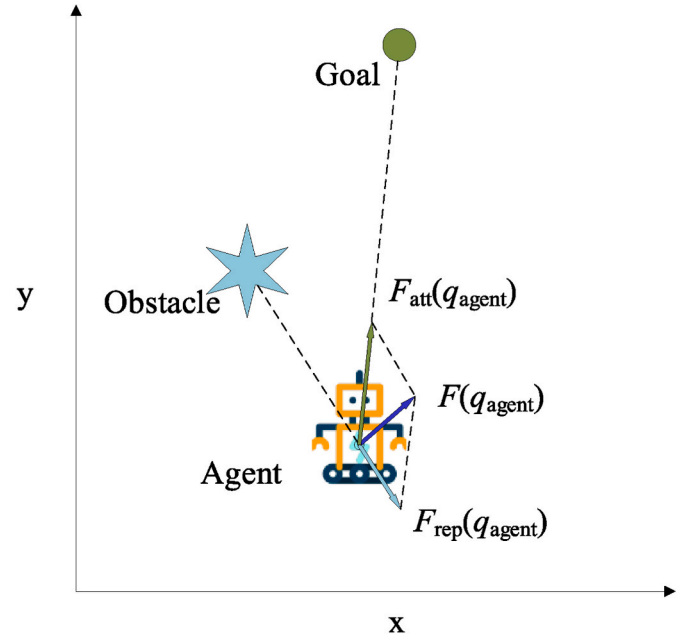


Fig. 6. Forces of APF-based mobile robot.

for s_t ; α_L is learning rate, which values between 0 and 1; γ is discount rate, which values between 0 and 1; ϵ is greed rate.

The likelihood that the agent chooses the action that owns the highest behavioral value is denoted as $1-\epsilon$; while the likelihood that the agent arbitrarily chooses the subsequent action to be executed by the mobile robot from all possible actions is indicated as ϵ . After many iterations, the Q-learning-based agent eventually obtains the optimal set of actions. The main problem of Q-learning is the slow convergence for complex environments. The planning process of a Q-learning-based mobile robot is illustrated in Fig. 7.

3.4. Fuzzy A* algorithm of fuzzy A* quantum multi-stage Q-learning artificial potential field

The A* algorithm obtains the optimal path from the start point to the goal point by calculating the heuristic function of nodes. In general, the greater the weight α_h of the A* algorithm, the less time the A* algorithm

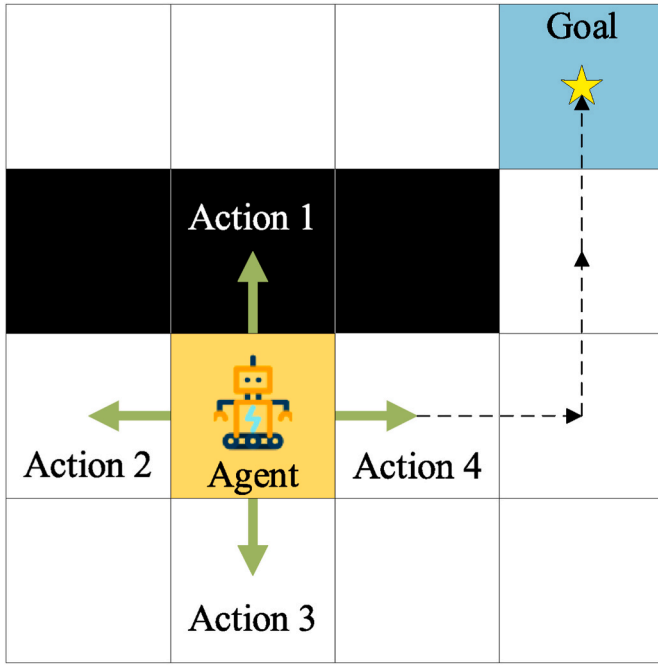


Fig. 7. Q-learning planned path.

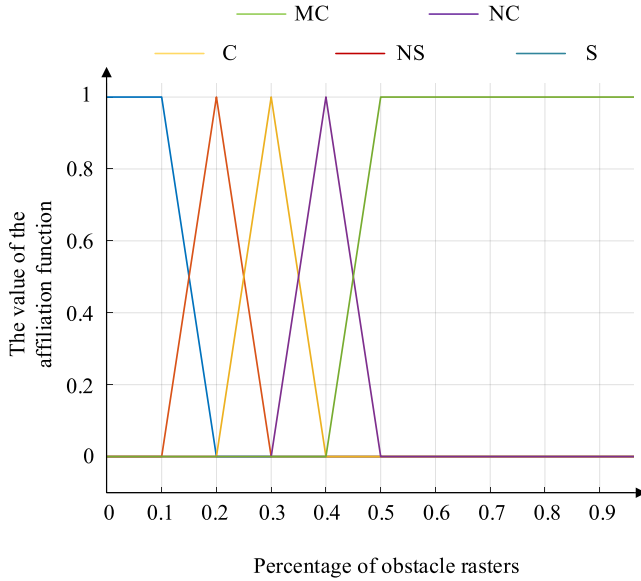


Fig. 8. Affinity function.

spends and the less reasonable the path is. On the contrary, the A* algorithm spends more time and the path is more reasonable.

When mobile robots are in a simple environment, the weight α_h should be large to speed up the planning of the A* algorithm. However, when mobile robots are in a complex environment, the weight α_h should be small to plan paths more rationally.

To balance the conflict between the time spent on paths and the length of paths, this study proposes the fuzzy A* algorithm. With α_g constant at 1, the fuzzy A* algorithm can adjust the weights of α_h in the heuristic function according to the complexity of the environment to make the A* algorithm more adaptable to environments.

The fuzzy A* algorithm fuzzifies the occupancy of obstacle rasters in raster environments into fuzzy quantities expressed linguistically by means of a fuzzy controller. The fuzzy subset of the occupancy of

obstacle rasters in the raster environment is defined as {MC (Most Complex), NC (Normal Complex), C (Complex), NS (Normal Simple), S (Simple)} (Yin and Li, 2022).

The affiliation function for the percentage of obstacle rasters in a raster environment consists of five triangular affiliation functions. Fig. 8 shows the affiliation function of the percentage of obstacle rasters in a raster environment. The definition of the affiliation function $F_{\text{Triangle}}(x, a_F, b_F, c_F)$ can be expressed as follows.

$$F_{\text{Triangle}}(x, a_F, b_F, c_F) = \begin{cases} 0, & x \leq a_F \text{ or } x \geq c_F \\ \frac{x - a_F}{b_F - a_F}, & a_F \leq x \leq b_F \\ \frac{c_F - x}{c_F - b_F}, & b_F \leq x \leq c_F \end{cases} \quad (10)$$

where x is the percentage of the obstacle grid; a_F and c_F are the coordinates of the intersection of the left and right hypotenuse of the triangle with the base edge, respectively; b_F are the coordinates of the triangle vertex.

The fuzzy A* algorithm adjusts α_f according to the output of the fuzzy controller. The weight α_f of the heuristic function is adjusted as follows.

$$\alpha_f = \begin{cases} 0, & \text{MC} \\ 0.5, & \text{NC} \\ 1, & \text{C} \\ 1.5, & \text{NS} \\ 2, & \text{S} \end{cases} \quad (11)$$

where F_{Triangle} is the output of the fuzzy controller.

3.5. Quantum multi-stage Q-learning algorithm of fuzzy A* quantum multi-stage Q-learning artificial potential field

Traditional Q-learning algorithms suffer from the dimensionality catastrophe problem. Therefore, when environments are large, the Q learning algorithm suffers a long convergence time. Besides, the path planning problem is a typical reward sparsity problem. As a result, the value of an action is challenges to determine until the agent has reached the goal point. In the reward sparsity problem, the training time of the Q-learning agents tends to be wasted a lot in the previous ineffective exploration.

This study proposes the quantum multi-stage Q-learning algorithm. The proposed quantum multi-stage Q-learning algorithm applies quantum computing methods and multi-stage training methods to improve the Q-learning algorithm. The proposed quantum multi-stage Q-learning algorithm can speed up the convergence of Q-learning agents.

The quantum multi-stage Q-learning algorithm represents the q-value corresponding to an action by means of a quantum bit probability amplitude. Therefore, the q-value of an action of a quantum multi-stage Q-learning agent can be a superposition of multiple states. In the quantum probability amplitude approach, the state of a quantum bit can generally be expressed as follows.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (12)$$

$$\alpha^2 + \beta^2 = 1, \quad (13)$$

where $|\psi\rangle$ is the state of a quantum bit; α and β are the quantum bit probability amplitudes of the 0 and 1 states, respectively; $|0\rangle$ is the 0 state of the quantum state; $|1\rangle$ is the 1 state of the quantum state (Yin and Cao, 2022).

Assuming that θ denotes the phase, let $\alpha = \cos \theta$ and $\beta = \sin \theta$. The quantum bit can be denoted as $[\cos \theta \quad \sin \theta]^T$, which can also be called the probability amplitude of the quantum bit.

The action of an agent in a state contains multiple quantum bits. The q-value of the action of an agent is expressed as follows.

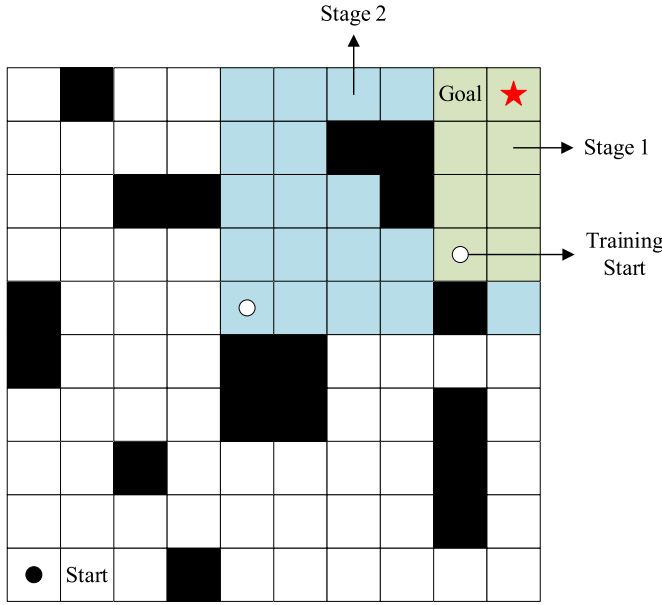


Fig. 9. Training process of quantum multi-stage Q-learning algorithm.

$$q = \begin{bmatrix} \cos(\theta_{j1}) & \cos(\theta_{j2}) & \dots & \cos(\theta_{jm}) \\ \sin(\theta_{j1}) & \sin(\theta_{j2}) & \dots & \sin(\theta_{jm}) \end{bmatrix}, \quad (14)$$

where q is the q-value of an action (Yin and Cao, 2024).

Since the quantum Q learning algorithm proposed in this study has four corresponding actions in a state. Therefore, two quantum bits are applied in this study to represent the actions corresponding to the states. The phases of the two quantum bits are set as θ_1 and θ_2 . Then θ_1 and θ_2 satisfy the following relationship.

$$\cos^2(\theta_1) + \sin^2(\theta_1) [\cos^2(\theta_2) + \sin^2(\theta_2)] = 1, \quad (15)$$

When the Q-learning algorithm converges, the quantum bits corresponding to the actions converge to 0 or 1, respectively. Compared to the traditional Q-learning algorithm, the actions of the quantum multi-stage Q-learning algorithm converge according to the square relationship. Therefore, the quantum multi-stage Q-learning algorithm can complete the convergence faster than the traditional Q-learning algorithm.

Besides, the fuzzy A*-quantum multi-stage Q-learning algorithm splits the training environment of agents from a large environment to a series of superpositions of small environments to reduce the unnecessary exploration of the agents.

First, the quantum multi-stage Q-learning algorithm segments the environment into mutually contained environments. Then, agents are trained sequentially according to the segmented environments.

Fig. 9 shows a schematic representation of the training process of the quantum multi-stage Q learning algorithm. The quantum multi-stage Q-learning algorithm first starts training from a region with a small area. The agent selects the raster in the training area that is far away from the target point and is not an obstacle raster as the starting point of the training process. Take the case of Fig. 9 as an example. At the first training process, the training process of the agent is restricted to the green area. When the agent has finished training in the green area, the agent retains the trained q table for subsequent training. The training area of the agent is restricted to the blue rectangular area and the green rectangular area. And so on, until the agent completes the training of the whole environment, the resulting q table is the final q table.

In the training process, when passing through the training region of the previous stage, agents can quickly reach the target point based on the previous q -value. Therefore, the quantum multi-stage Q-learning algorithm can reduce unnecessary exploration to speed up the convergence.

The quantum multi-stage Q-learning algorithm proposed in this study has two modes: local path planning and global path planning.

The state obs of the quantum multi-stage Q learning algorithm in local path planning mode can be represented as follows.

$$obs = [S, D_X, D_Y], \quad (16)$$

where, S is the obstacle information around the agent; D_X is the horizontal coordinate deviation of the position of agent from the position of target point; D_Y is the vertical coordinate deviation of the position of agent from the position of target point.

In the local path planning mode, the agent can apply the trained Q-table to plan paths in different environments without the need of training again and has the advantage of high real-time performance. In the local path planning mode, the more information about the obstacles around the agent in the agent state volume, the better the ability of agent to plan paths. However, too much state volume will cause the Q learning algorithm to suffer from the dimension explosion problem resulting in challenge in training. Therefore, consistent with other local path planning algorithms, the Q-learning algorithm applying local path planning model can fail to plan a path because the observation magnitude problem. Therefore, this study sets both local and global modes of Q learning. When the Q-learning algorithm applying the local mode fails to plan a path, the Q-learning algorithm switches to the global mode.

In the global mode, the agents need to be trained for different environments resulting in poor real-time performance. However, the Q learning algorithm applying global mode belongs to the probabilistic complete path planning algorithm. Therefore, the Q learning algorithm applying global mode can plan paths in complex environments.

In global mode, the state of the quantum multi-stage Q-learning algorithm is the position of agent and the position of the endpoint.

For the purposes of comparison with other Q-learning algorithms, all training time comparisons are conducted in the regular global mode.

In this study, the output action of the quantum multi-stage Q-learning algorithm is a 4-direction movement including front, back, left, and right of the location of agent.

3.6. Fuzzy A* quantum multi-stage Q-learning APF

The proposed fuzzy A* quantum multi-stage Q-learning APF combines the fuzzy A*, the APF, and the quantum multi-stage Q-learning (Fig. 10 and Algorithm 1). The fuzzy A* quantum multi-stage Q-learning APF employs the fuzzy A* algorithm to plan a global path. If dynamic obstacles exist in environments and block the global path, the APF utilizes the global path nodes, which have been removed as redundant nodes, as sub-goal points to plan a path. Such sub-goal points are referred to as A* sub-goal points.

The A* algorithm is a complete path planning algorithm and has the advantage of fast planning speed. Therefore, the A* algorithm is applied as the global path planning algorithm in this study. The following presents the approach for removing redundant nodes from the global path. This method designates the current node as the first node, followed by sorting subsequent nodes sequentially. A connection line is then created between the first and third nodes. If this line intersects with an obstacle, there is no need to eliminate redundant nodes from the first node. However, if the line is no such intersection, the first node is connected to the next one. Until the first node is connected to a node through an obstacle, such a node is called a non-directly reachable node. Meanwhile, the previous node of the non-directly reachable node is called a directly reachable (DR) node. The connection from the first node to the DR node is a new path segment. Subsequently, the DR node is selected as the new first node. Repeat the above process until all redundant nodes have been removed.

Although the path planned by the A* algorithm without removing redundant nodes can also be applied to guide the mobile robot employing the APF algorithm to escape from the trap region, frequent

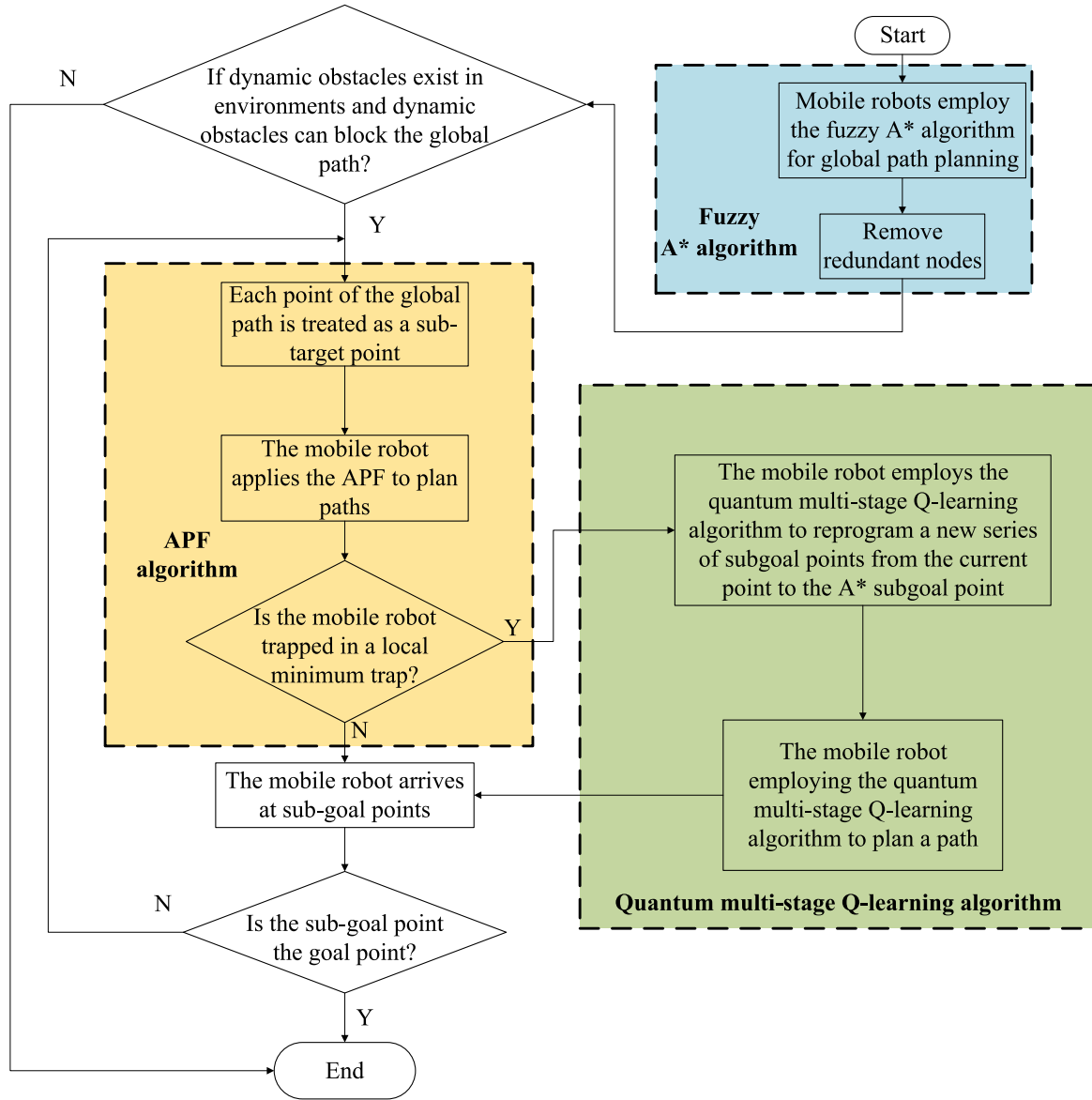


Fig. 10. Flow chart of fuzzy A* quantum multi-stage Q-learning APF.

switching of sub-target points can lead to a greater degree of path curvature for the mobile robot employing the APF algorithm. Therefore, the path with redundant nodes removed can smooth the planned path.

If dynamic obstacles exist in environments and dynamic obstacles can block the global path, a mobile robot plans local paths between each sub-goal point in turn by APF. When an A* sub-goal point is reached, the goal point of APF is switched to the next A* sub-goal point. The A* algorithm is the complete path planning algorithm. With accurate environment information, the A* algorithm can plan a collision-free global path. Therefore, the path planned by the A* algorithm cannot fall into the trap region making the A* subgoal point can guide the mobile robot that applies the APF algorithm to plan the path to bypass the trap region. By combining the fuzzy A* algorithm and APF, the advantages of the fuzzy A* algorithm and the APF are exploited. Specifically, the APF can solve the problem that the fuzzy A* algorithm cannot dynamically avoid obstacles; the fuzzy A* algorithm can provide global information to the APF process of the mobile robot.

For hybrid algorithms that combine the A* algorithm with the APF,

most previous literature tends to simply assume that the APF can effectively keep mobile robots away from local minimum traps when guided by sub-goal points. However, most existing literature had not an effective solution for mobile robots to escape from local minimum traps. Nevertheless, during movement, an APF-based mobile robot may deviate from the global path because of avoiding dynamic obstacles. After the mobile robot has deviated from the global path, the guiding effect of the A* algorithm on the APF process does not enable the mobile robot to bypass obstacle regions. Moreover, even with the guidance of the A* algorithm, APF-based mobile robots still face difficulties when passing through narrow passages. None of the mentioned problems of the APF can be solved by the guidance of A* sub-goal points. Although coefficient transformations can partially solve the above problems, previous improvements have often failed to effectively address the above issues. Therefore, this study introduces quantum multi-stage Q-learning to solve the above problems. When the position of a mobile robot planning a path applying the APF algorithm changes regularly back and forth in several fixed coordinates, the mobile robot falls into a

local minimum trap. When caught in a local minimum trap, a mobile robot invokes quantum multi-stage Q-learning to re-plan a path to reach the A* sub-goal point.

After being trapped in a local minimum trap, a mobile robot may escape by following a series of quantum multi-stage Q-learning re-planned sub-goal points. Such re-planned sub-goal points are called the Q sub-goal points. The Q sub-goal points start at the current point, where the mobile robot is caught in trap areas, and end at the A* sub-goal point that should be reached.

When quantum multi-stage Q-learning for path planning, information is obtained continuously by interacting with environments. The mobile robot obtains a path from trap areas to the A* sub-goal point after sufficient learning. Then, the mobile robot reaches each Q sub-goal in turn. During movement, the mobile robot continuously detects whether an obstacle appears at the next path point. If an obstacle appears in the path, the mobile robot re-plans paths.

Although quantum multi-stage Q-learning is applied as a local path-planning algorithm that can achieve dynamic obstacle avoidance and escape from local minimum traps, the paths planned by quantum multi-stage Q-learning have a large degree of curvature. Therefore, this study applies the quantum multi-stage Q-learning for local path planning as an emergency planning algorithm rather than an entire path planning algorithm when the APF is unable to perform path planning.

Algorithm 1. Steps of fuzzy A* quantum multi-stage Q-learning APF.

Input: environment information, goal point, start point
Output: path nodes
Initialize: the parameters of the formula, Q table
1. Obtain the global path through the fuzzy A* algorithm
2. If dynamic obstacles exist in environments and dynamic obstacles can block the global path do
3. flag = 1
4. end if
5. while flag do
6. for each A* sub-goal do
7. while the position of a mobile robot $\neq$ A* sub-goal do
8. Plan a path to the next A* sub-goal through APF
9. if the mobile robot falls into a trap area do
10. Plan a path to the next A* sub-goal through quantum multi-stage Q-learning
11. end if
12. end while
13. end for
14. end while

Table 2
Parameters of the compared algorithms.

Algorithm	Parameter	Range
All algorithms	Movement step	0.1
A*	The weights of heuristic functions	1
APF	The gain coefficient of the attractive force	4
APF	The gain coefficient of repulsive force	15
APF	The distance that the repulsive force can affect	5
Q-learning	Greed rate	0.2
Q-learning	Learning rate	0.4
Q-learning	Discount rate	0.7
QAPF	Greed rate	0.2
QAPF	Learning rate	0.4
QAPF	Discount rate	0.7
Quantum multi-stage Q-learning	Greed rate	0.2
Quantum multi-stage Q-learning	Learning rate	0.4
Quantum multi-stage Q-learning	Discount rate	0.7
Fuzzy A* quantum multi-stage Q-learning APF	The maximum of conversions	2

4. Results and analysis

This study employs MATLAB R2020a as software on a computer running Windows 10 with a CPU model of AMD R5 3600 (3.6 GHZ) and 16 GB of RAM.

To verify the effectiveness of the quantum multi-stage Q-learning algorithm proposed in this study in reducing the convergence time, a comparison is made between the planning time of the proposed quantum multi-stage Q-learning algorithm, the QAPF algorithm, the traditional Q-learning algorithm, and quantum multi-stage Q-learning for planning paths in a randomized environment.

To test the performance of the fuzzy A* quantum multi-stage Q-learning APF for complex environments, this study compares the results of the proposed fuzzy A* quantum multi-stage Q-learning APF with the A* algorithm, APF, Q-learning, and A*-APF. The designed five cases are a simple random environment without dynamic obstacles, a complex random environment without dynamic obstacles, a non-random environment with trap obstacles, a non-random environment with a narrow channel, and a complex environment. Since the fuzzy A* quantum multi-stage Q-learning APF is a fusion of the fuzzy A*, APF, and quantum multi-stage Q-learning, the parameters of the A*, APF, and quantum multi-stage Q-learning are the parameters of the fuzzy A* quantum multi-stage Q-learning APF. Table 2 demonstrates the parameter values for different algorithms.

4.1. Comparison of planning time for Q-learning and improved algorithms

To verify the effectiveness of the quantum multi-stage Q-learning algorithm proposed in this study in terms of reducing the convergence time, a comparison is made between the proposed quantum multi-stage Q-learning, the QAPF (QAPF algorithm utilizes the APF algorithm to provide the Q-learning algorithm with the a priori knowledge for the training process), and the traditional Q-learning in terms of the time taken to plan paths in the same environment.

The sizes of the randomized environments set up in this study are 20×20 , 30×30 , and 40×40 . The percentage of obstacle raster for all environments is 10%.

Fig. 11 shows the randomly generated raster environment. Table 3 shows the convergence time of different algorithms in the raster environment of Fig. 11. From Table 3, the planning time of the quantum multi-stage Q-learning algorithm proposed in this study is less compared to the traditional Q-learning algorithm and QAPF algorithm. Compared to the traditional Q-learning algorithm, the proposed quantum multi-stage Q-learning algorithm can reduce the convergence time by 80.54% on average. Therefore, the quantum multi-stage Q-learning proposed in this study can effectively reduce convergence time.

4.2. Simple randomized environment without dynamic obstacles (Case 1)

The average distance between obstacles of Case 1 is relatively large. In Case 1, trap areas and narrow passages are not easily formed to impede the passage of mobile robots.

The planned paths are presented in Fig. 12. The planning time of comparison algorithms in Case 1 is presented in Table 4. The fuzzy A* quantum multi-stage Q-learning APF plans the shortest path and has minimal consumption of time in Case 1. The performance of the A* algorithm is almost the same as fuzzy A* quantum multi-stage Q-learning APF in Case 1. A mobile robot with the APF does not fall into local minimum traps. A mobile robot with Q-learning has only four selectable actions in a state. Therefore, the path planned by Q-learning is curved while avoiding the path overly close to obstacles. In terms of time consumption, the quantum multi-stage Q-learning algorithm proposed in this study takes 27.76% of the time applied by the Q-learning algorithm to plan a path. However, the path planned by the quantum multi-stage Q-learning algorithm also suffers from a greater degree of path bending. The A*-APF can plan a collision-free path, while the planning

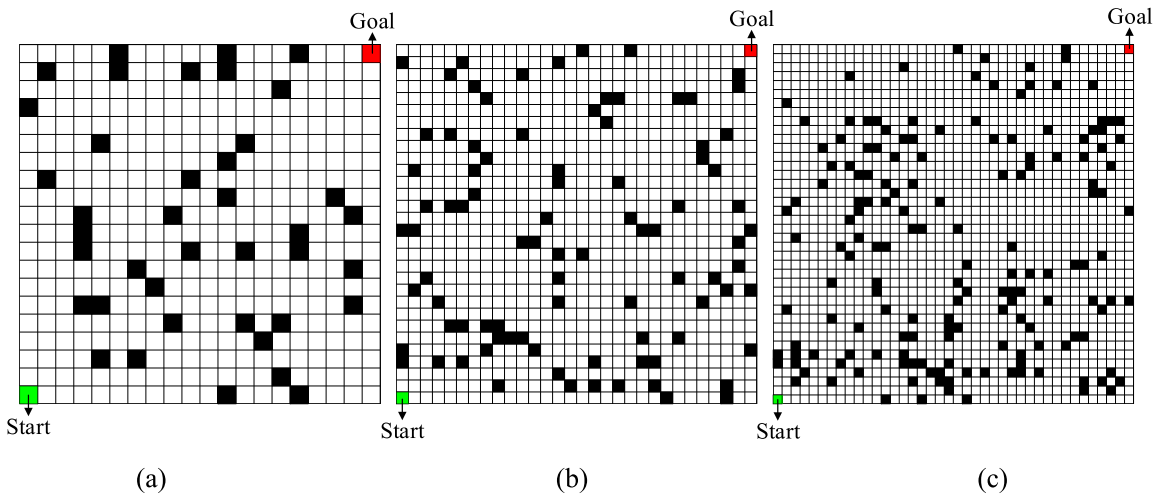


Fig. 11. Randomly generated environments: (a) environment 1; (b) environment 2; (c) environment 3.

Table 3
Comparison of convergence times of different algorithms.

Algorithm	Environment 1 (s)	Environment 2 (s)	Environment 3 (s)
Q-learning	3.44	9.53	21.49
QAPF	0.90	2.86	7.31
Quantum multi-stage Q-learning	0.66	1.86	4.23

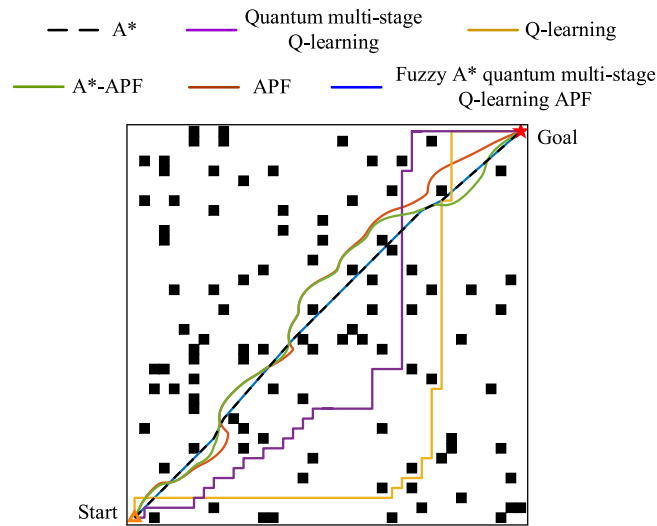


Fig. 12. Planned paths of different algorithms for Case 1.

Table 4
Comparison of algorithms in Case 1.

Algorithm	Reach the goal point?	Planning time (s)	Length of path (p.u.)
A*	Yes	0.24	55.43
APF	Yes	0.39	60.23
Q-learning	Yes	22.37	78.00
Quantum multi-stage Q- learning	Yes	6.21	78.00
A*-APF	Yes	1.69	63.20
Fuzzy A* quantum multi- stage Q-learning APF	Yes	0.21	55.43

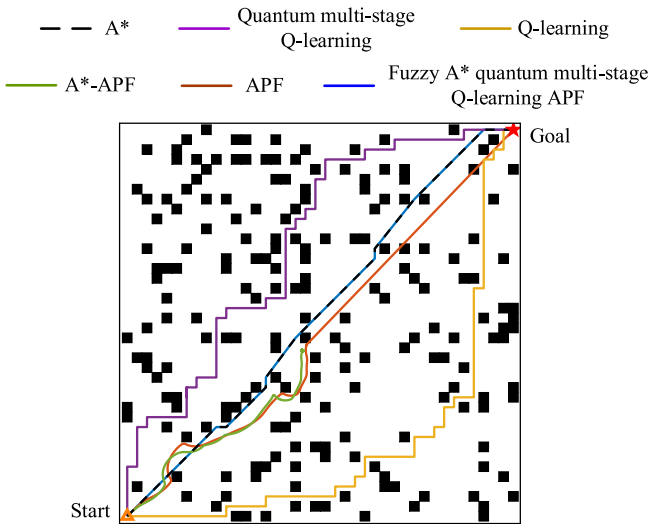


Fig. 13. Planned paths obtained by compared algorithms in Case 2.

Table 5
Comparison of algorithms in Case 2.

Algorithm	Reach the goal point?	Planning time (s)	Length of path (p.u.)
A*	Yes	0.28	57.49
APF	No	0.41	29.10
Q-learning	Yes	23.01	78.00
Quantum multi-stage Q- learning	Yes	6.39	78.00
A*-APF	No	0.76	28.94
Fuzzy A* quantum multi- stage Q-learning APF	Yes	0.26	57.49

time is consumed more than fuzzy A* quantum multi-stage Q-learning APF.

Mobile robots applying the APF are not easily trapped in local minimum traps in Case 1. Paths planned by fuzzy A* quantum multi-stage Q-learning APF and A* algorithm overlap when mobile robots are not trapped in local minima traps and without the effect of dynamic obstacles. Path overlap occurs because the proposed fuzzy A* quantum multi-stage Q-learning APF does not apply the quantum multi-stage Q-learning of fuzzy A* quantum multi-stage Q-learning APF and the APF of

fuzzy A* quantum multi-stage Q-learning APF when mobile robots are not caught in the local minima trap and without the effect of dynamic obstacles. The fuzzy A* quantum multi-stage Q-learning APF merely applies the fuzzy A* algorithm as a global path planning algorithm. Therefore, the planned path of the fuzzy A* quantum multi-stage Q-learning APF is almost consistent with the A* algorithm.

4.3. Intricate randomized environment without dynamic obstacles (Case 2)

In Case 2, average distances between different obstacles are relatively small; trap areas and narrow passages can easily be formed to impede the passage of mobile robots.

Fig. 13 shows the paths planned by different algorithms in Case 2. The planning time of comparison algorithms in Case 2 is presented in Table 5. Even in a complex environment, a collision-free path can be found by the A* algorithm as a completeness planning algorithm. The Q-learning can achieve dynamic obstacle avoidance for mobile robots. However, planning time is severely consumed for Q-learning in such a large environment. The quantum multi-stage Q-learning algorithm proposed in this study takes 27.77% of the time applied by the Q-learning algorithm to plan a path. In Case 2, the APF is susceptible to being trapped in local minimum traps. Both a mobile robot with APF and a mobile robot with A*-APF fall into a trap area at the halt point. Table 5 shows that the performance of the A* algorithm is almost the same as fuzzy A* quantum multi-stage Q-learning APF in Case 2 when dynamic obstacles do not exist in environments. Thus, the fuzzy A* quantum multi-stage Q-learning APF can plan the shortest collision-free path with the least time consumption.

4.4. Environment with trap obstacles (Case 3)

To verify if the proposed fuzzy A* quantum multi-stage Q-learning APF can avoid mobile robots falling into local minimum traps when dynamic obstacles do not exist in environments, two different environments are designed in this case. The first environment consists of an environment without any dynamic obstacles and is designed to evaluate the efficacy of the A* algorithm in guiding mobile robots away from a local minimum trap. The second environment contains dynamic obstacles and aims to test the ability of the A*-APF and fuzzy A* quantum multi-stage Q-learning APF to escape from local minimum traps.

The planning time of comparison algorithms in Case 3 without dynamic obstacles is presented in Table 6; the planned paths are presented in Fig. 14 (a). A* algorithm-based mobile robot can quickly find a path in Case 3 without dynamic obstacles. The APF-based mobile robot alone does not have sufficient global information, and then falls into a trap region. However, the A* sub-goal points generated by the A* algorithm of A*-APF can guide the mobile robot to bypass trap areas. The path planned by the fuzzy A* quantum multi-stage Q-learning APF should be almost consistent with the path planned by the A* algorithm when mobile robots are without the effect of dynamic obstacles. The fuzzy A* quantum multi-stage Q-learning APF needs the least time consumption. In summary, the above simulation results demonstrate that the A*

Table 6
Comparison of algorithms in Case 3 without dynamic obstacles.

Algorithm	Reach the goal point?	Planning time (s)	Length of path (p.u.)
A*	Yes	0.16	28.68
APF	No	0.08	10.20
Q-learning	Yes	3.62	38.00
Quantum multi-stage Q-learning	Yes	1.10	38.00
A*-APF	Yes	0.76	29.40
Fuzzy A* quantum multi-stage Q-learning APF	Yes	0.15	28.68

algorithm can guide a mobile robot applying the APF to bypass part of trap areas, thus reducing the number of cases where a mobile robot gets caught up in local minimum traps.

The planning time of comparison algorithms in Case 3 with dynamic obstacles is presented in Table 7; the planned paths are presented in Fig. 14(b). A* algorithm-based mobile robot cannot find a path because of the influence of dynamic obstacles. Thus, fuzzy A* quantum multi-stage Q-learning APF applies the APF of fuzzy A* quantum multi-stage Q-learning APF to plan a path between A* sub-goal points. The APF-based mobile robot falls into local minimum traps because of lacking global information. Q-learning-based mobile robots can reach the goal point in Case 3. Besides, the time of Q-learning cost is significantly less than in Case 1 and Case 2 because the exploration area is small. The time consumption of the quantum multi-stage Q-learning algorithm proposed in this study is 30.39% and 30.13% of that of the Q-learning algorithm in environments that do not contain dynamic obstacles and environments that contain dynamic obstacles, respectively. Although sub-goal points generated by the A* algorithm can guide mobile robots bypasses some trap areas, the mobile robots may still get caught up in the trap areas during obstacle avoidance. However, the A*-APF cannot escape from the trap areas. To escape from trap areas, the quantum multi-stage Q-learning with a strong optimization-seeking capability of fuzzy A* quantum multi-stage Q-learning APF can reprogram the Q sub-goal points to escape the trap area and reach the A* sub-goal point.

In summary, APF-based mobile robots are prone to fall into local minimum traps. Because of A* algorithm can provide global information to the APF, the cases in which APF-based mobile robots fall into local minimum trap regions are reduced. However, the A*-APF-based mobile robots cannot escape from trap areas when mobile robots are caught in local minimum traps because of dynamic obstacles. In contrast, the proposed fuzzy A* quantum multi-stage Q-learning APF can guide the APF process and effectively solves the problem of mobile robots escaping from local trap areas.

4.5. Environment with narrow passages (Case 4)

In this section, Case 4 is designed to verify the ability of a fuzzy A* quantum multi-stage Q-learning APF-based mobile robot to pass narrow pathways when a dynamic obstruct exists.

Fig. 15 shows the paths planned by different algorithms in Case 4. The comparison of algorithms in Case 4 is presented in Table 8. A mobile robot applying the A* algorithm can quickly find a global path in Case 4. However, the A* algorithm-based mobile robot cannot reach the goal point because dynamic obstructs block the path. A mobile robot applying the APF cannot pass narrow pathways. A mobile robot applying the Q-learning can through narrow passages. The time consumption of the mobile robot planning a path applying the quantum multi-stage Q-learning algorithm is 25.80% of the time consumption of the mobile robot planning a path applying the Q-learning algorithm. Mobile robots with the A*-APF fall into a local minimum trap in front of the narrow passage. The reasons why the mobile robots get caught up in the trap area include the fact that the mobile robots are subject to a large repulsive force in front of the narrow passage and the fact that the closer the goal point is to the mobile robots, the less attractive the goal point generated. Hence, although the guidance of A* sub-goal point can guide mobile robots to bypass some obstacle areas, the guidance of A* sub-goal point still cannot solve the problem of the mobile robot not being able to pass through the narrow passage.

To through the narrow passages, the mobile robot relies on the Q-learning for local path planning. Simulation results showed that the Q-learning enabled the mobile robot to successfully through narrow pathways. Besides, the time consumption of fuzzy A* quantum multi-stage Q-learning APF is less than Q-learning. The reason why the mobile robot applying the fuzzy A* quantum multi-stage Q-learning APF can pass through the narrow passage and have less time consumption is that the fuzzy A* quantum multi-stage Q-learning APF proposed in this

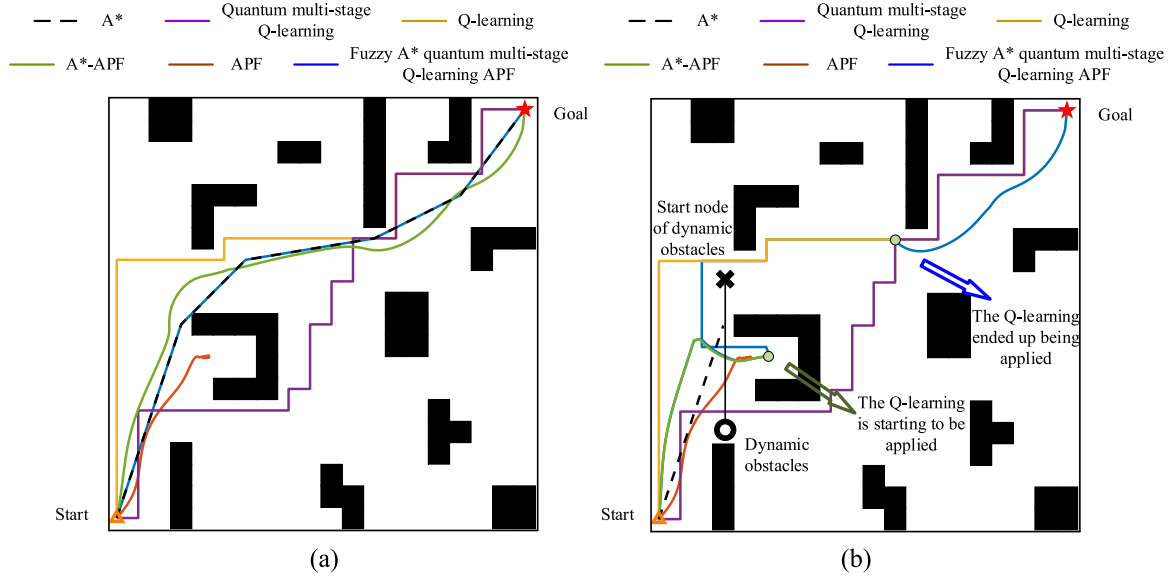


Fig. 14. Dynamic obstacle avoidance and escape from local minimum traps for mobile robots: (a) without dynamic obstacles; (b) with dynamic obstacles.

Table 7
Comparison of algorithms in Case 3 with dynamic obstacles.

Algorithm	Reach the goal point?	Planning time (s)	Length of path (p.u.)
A*	No	0.11	8.55
APF	No	0.10	10.20
Q-learning	Yes	3.85	38.00
Quantum multi-stage Q-learning	Yes	1.16	38.00
A*-APF	No	0.29	13.50
Fuzzy A* quantum multi-stage Q-learning APF	Yes	0.87	39.25

Table 8
Comparison of algorithms in Case 4.

Algorithm	Reach the goal point?	Planning time (s)	Length of path (p.u.)
A*	No	0.12	8.64
APF	No	0.09	14.10
Q-learning	Yes	3.45	38.00
Quantum multi-stage Q-learning	Yes	0.89	38.00
A*-APF	No	0.25	13.20
Fuzzy A* quantum multi-stage Q-learning APF	Yes	0.67	31.45

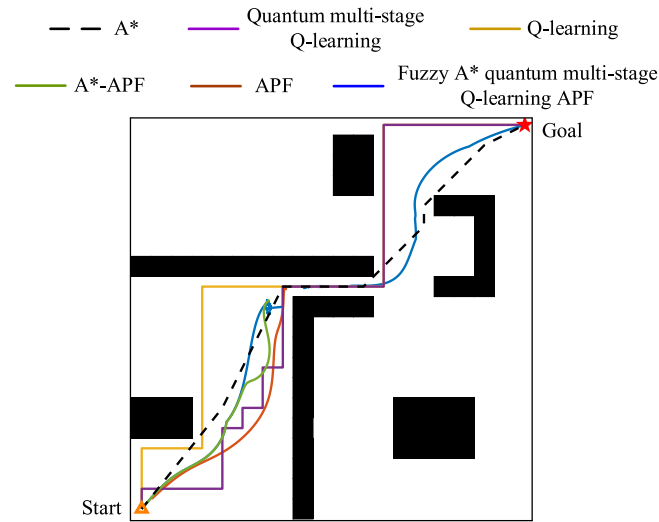


Fig. 15. Mobile robot through narrow passages.

study combines the advantages of the A*, the APF, and the Q-learning, therefore has a strong merit-seeking capability. Thus, fuzzy A* quantum multi-stage Q-learning APF can utilize the Q-learning algorithm to pass through the narrow channel and reach the A* next subgoal point.

In summary, a mobile robot applying the fuzzy A* quantum multi-stage Q-learning APF proposed in this study can effectively pass

through narrow passages.

4.6. Complex environment (Case 5)

In this section, a complex environment is designed to validate the ability of the proposed fuzzy A* quantum multi-stage Q-learning APF algorithm to plan paths in complex environments. The obstacles of the designed complex environment contain static and dynamic obstacles.

Fig. 16 shows the designed complex environment and the paths planned by each algorithm. Table 9 shows the comparison of the length and time consumption of the planned paths of mobile robots.

In complex environments, the A* algorithm cannot handle dynamic obstacles in the environment. As a result, the mobile robot applying the A* algorithm cannot reach the target point. Mobile robots applying the APF algorithm are prone to fall into trap areas in the environment. Therefore, the mobile robot applying the APF algorithm for path planning cannot reach the goal point. The mobile robot applying Q-learning algorithm can reach the goal point. However, the mobile robot that applies Q-learning algorithm to plan the path takes more time and the path is not smooth enough. Compared to the traditional Q-learning algorithm, the time consumption of the quantum multi-stage Q-learning algorithm proposed in this study is 22.33%. Similarly, the paths planned by the quantum multi-stage Q learning algorithm are not smooth enough. The mobile robot planning path applying the A*-APF algorithm still cannot overcome the disadvantage of the APF algorithm that easily falls into the local minimum trap region.

The mobile robot applying the fuzzy A* quantum multi-stage Q-learning APF algorithm proposed in this study improves the path-finding ability of the algorithm by combining the advantages of the A*

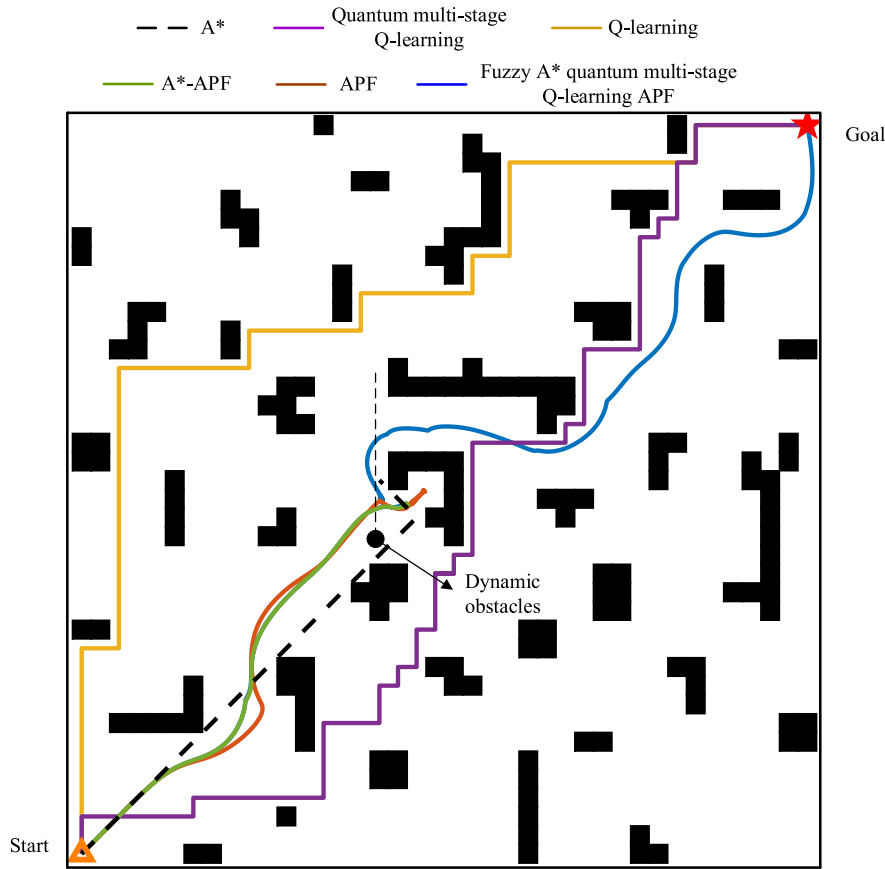


Fig. 16. Different algorithms for planning paths in complex environments.

Table 9
Comparison of algorithms in Case 5.

Algorithm	Reach the goal point?	Planning time (s)	Length of path (p.u.)
A*	No	0.41	22.63
APF	No	0.49	30.70
Q-learning	Yes	24.99	78.00
Quantum multi-stage Q-learning	Yes	5.58	78.00
A*-APF	No	3.30	28.70
Fuzzy A* multi-staged quantum Q-learning APF	Yes	1.36	59.90

However, this study has some deficiencies that are summarized as follows.

algorithm, the APF algorithm, and quantum multi-stage Q-learning. When mobile robots are caught in the local minimum trap region, the mobile robots can apply the quantum multi-stage Q-learning algorithm to re-plan the sub-goal points to help the mobile robots escape from the local minimum trap region. Therefore, the mobile robot applying the fuzzy A* quantum multi-stage Q-learning APF algorithm proposed in this study can successfully plan the path of the mobile robot when facing a complex environment.

In summary, the fuzzy A* quantum multi-stage Q-learning APF algorithm proposed in this study is able to handle dynamic obstacles in the environment and escape from the local minimum trap region.

- 1) The optimality of a path planned by the fuzzy A* quantum multi-stage Q-learning APF cannot be ensured. Because of the influence of various obstacles, the proposed fuzzy A* quantum multi-stage Q-learning APF-based mobile robot may deviate from a global path in the local path-planning process. Therefore, a path planned by the

fuzzy A* quantum multi-stage Q-learning APF may not be optimal for complex environments with multiple obstacles.

- 2) Although the proposed fuzzy A* quantum multi-stage Q-learning APF algorithm can plan paths of mobile robots in complex environments, the paths planned by the fuzzy A* quantum multi-stage Q-learning APF algorithm are not sufficiently compliant with the kinematic constraints of mobile robots.

5. Conclusions

Combining the fuzzy A* algorithm, APF, and quantum multi-stage Q-learning, the fuzzy A* quantum multi-stage Q-learning APF is proposed to address path-planning problems under complex environments in this study. The fuzzy A* quantum multi-stage Q-learning APF balances global and local path planning. Moreover, the fuzzy A* quantum multi-stage Q-learning APF solves the problem that mobile robots are easy to fall into local minimum traps. The proposed fuzzy A* quantum multi-stage Q-learning APF has the following features.

- 1) Compared with the A* algorithm, the proposed fuzzy A* quantum multi-stage Q-learning APF can achieve dynamic obstacle avoidance. Compared with the Q-learning, the proposed fuzzy A* quantum multi-stage Q-learning APF does not have the problem of long convergence time when performing path planning in complex environments. Compared with the Q-learning, the proposed fuzzy A* quantum multi-stage Q-learning APF proposed in this study reduces the path curvature and consumes less time. Compared with the APF and A*-APF, the proposed fuzzy A* quantum multi-stage Q-learning APF can effectively guide mobile robots away from trap areas and local minimum traps.

- 2) If any dynamic obstacles block the global path, the fuzzy A* algorithm of fuzzy A* quantum multi-stage Q-learning APF can rapidly plan a globe path and provide globe information to the APF of fuzzy A* quantum multi-stage Q-learning APF. Thus, the APF of fuzzy A* quantum multi-stage Q-learning APF can bypass some obstacles that can trap mobile robots. Moreover, the APF of fuzzy A* quantum multi-stage Q-learning APF can provide the ability for dynamic obstacle avoidance. When the mobile robots get caught up in a trap area, the quantum multi-stage Q-learning of fuzzy A* quantum multi-stage Q-learning APF can escape from the trap area.
- 3) The proposed fuzzy A* quantum multi-stage Q-learning APF combines the advantages of the fuzzy A* algorithm that can rapidly plan a global path, the APF algorithm that can have dynamic obstacle avoidance, and quantum multi-stage Q-learning that can effectively respond to complex environments. Thus, the proposed fuzzy A* quantum multi-stage Q-learning APF has a stronger merit-finding capability and less time consumption to plan a path. In Case 1 and Case 2, the fuzzy A* quantum multi-stage Q-learning APF-based mobile robot can plan the shortest path with less time consumption.

In future works, (i) the combinatorial framework of the fuzzy A* quantum multi-stage Q-learning APF could be improved for reducing path length; (ii) an improved algorithm can satisfy kinematic constraints

for mobile robots; (iii) with the increasing importance of multi-mobile robot collaboration, the proposed algorithm in this study could be extended for multi-mobile robot path planning.

CRediT authorship contribution statement

Likun Hu: Supervision, Resources. **Chunyou Wei:** Writing – review & editing, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Linfei Yin:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Variables

$ 0\rangle$:	The 0 state of the quantum state
$ 1\rangle$:	The 1 state of the quantum state
α :	Quantum bit probability amplitudes of the 0 state
α_g :	Coefficient of $g(n)$
α_h :	Coefficient of $h(n)$
α_L :	Learning rate
β :	Quantum bit probability amplitudes of the 1 state
ε :	Greed rate
D_X :	the horizontal coordinate deviation of the position of agent from the position of target point
D_Y :	the vertical coordinate deviation of the position of agent from the position of target point
e_x :	Horizontal coordinate of the goal
e_y :	Vertical coordinate of the goal
$F(q_{\text{agent}})$:	Combined force on the mobile robot
$F_{\text{att}}(q_{\text{agent}})$:	Attractive force function
$F_{\text{rep}}(q_{\text{agent}})$:	Repulsive force function
$F_{\text{Triangle}}(x, a_F, b_F, c_F)$:	Affiliation function
γ :	Discount rate
K_a :	Gain coefficient of attractive force
K_r :	Gain coefficient of repulsive force
$L_1(n)$:	Manhattan distance calculation function
$L_2(n)$:	Euclidean distance calculation function
n_x :	Horizontal coordinate of the current node
n_y :	Vertical coordinate of the current node
$ \psi\rangle$:	State of a quantum bit
q :	Q-value of an action
q_{agent} :	Coordinates of the current point
$\rho(q_{\text{agent}}, q_{\text{goal}})$:	Distance between current point and goal
$\rho(q_{\text{agent}}, q_{\text{obs}})$:	Distance between the mobile robot and the obstacle
ρ_d :	Distance threshold
R_t :	Immediate return to the current state
S :	the obstacle information around the agent
s_t :	Current state
s_{t+1} :	Next state
θ :	Phase of a quantum bit
$U_{\text{att}}(q_{\text{agent}})$:	Function of the attractive potential field

$U_{rep}(q_{agent})$ Function of the repulsive potential field

Abbreviations

ABC	Artificial bee colony
APF	Artificial potential field
DR	Directly reachable
PSO	Particle swarm optimization
RRT	Rapidly exploring random tree

Data availability

Data will be made available on request.

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